

Portfolio Management (4. Alpha)

Active Portfolio Management

- In portfolio management, active management is all about performance relative to a benchmark. A few important parameters are defined on excess return over a suitable benchmark

Excess return (Active Return)

$$r_t^{ex} = r_t - r_t^{bmk}$$

Alpha (avg excess return)

$$\alpha = \frac{1}{T} \sum_{t=1}^T r_t^{ex}$$

$$\alpha = \frac{E[r_p] - r_f}{\sigma}$$

Tracking Error

TE

$$\sigma = \text{stdev}(r_t^{ex})$$

Information Ratio

$$IR = \frac{\alpha}{\sigma}$$

(coincides with Sharpe Ratio if benchmark = risk-free rate)

Benchmarks matter

- ▶ Alpha is often interpreted as manager's skill. But first and foremost, it's a statement about the benchmark. When we perform a CAPM style regression to obtain alpha, beta etc. the benchmark is not the market but a risk-adjusted market portfolio

$$\underline{R_{i,t}} - \underline{R_f} = \underline{\alpha} + \underline{\beta}(\underline{R_{m,t}} - \underline{R_f}) + \underline{\epsilon_{i,t}}$$

Rewrite the above equation

$$\underline{R_{i,t}} = \underline{\alpha} + \underbrace{[\underline{\beta R_{m,t}} + \underline{(1 - \beta)R_f}]}_{\text{(risk-adjusted benchmark)}} + \underline{\epsilon_{i,t}}$$

The benchmark is a blended portfolio of risk-free asset and the market. It will be wrong to conclude the benchmark as just market.

- ▶ An ideal benchmark must be Well-defined, Tradeable, Replicable, Adjusted for Risk.

Creating Alpha

- ▶ A portfolio manager creates Alpha relative to a benchmark by taking bets that deviate from that benchmark in terms of security selection and allocation.
- ▶ Grinold's fundamental law of active management states that the maximum Information ratio attainable is given by

$$\text{IR} \approx \text{IC} \times \sqrt{\text{BR}}$$

(Information coefficient)

(Breadth of strategy)

Correlation of actual
returns and manager's
forecasted returns

of bets and how frequently
one can trade them



Exercise - What is the IC needed to get an information ratio of 0.5 if a fund manager is allowed to take only 4 bets per year?

$$0.5 = \text{IC} \times \sqrt{4} \Rightarrow \text{IC} = 0.25$$

Assumptions of Grinold's Fundamental Law

- ▶ Assumes mean-variance utility framework (ignores asymmetry and higher moments risk)
- ▶ ICs are assumed to be constant across BR. In reality, IC of a manager diminishes as assets under management increases. ↔
- ▶ Bets in BR are independent of each other. In reality, the managers decisions become correlated and correlated bets reduce BR. An equity manager with overweight positions of 1000 value stocks and underweight positions on 1000 growth stocks has not put 1000 different bets, but it's a single bet on the value-growth factor.

$$IR = IC \times \sqrt{BR}$$

Diagram illustrating the components of the equation:

- IC (Information Coefficient) is circled in red.
- BR (Bets Return) is circled in red, with an arrow pointing to the word "bets" also circled in red.

Factor Benchmarks

- Consider the CAPM equation

$$\mathbb{E}[R_i] = R_f + \beta(\mathbb{E}[R_m] - R_f)$$

It can be re-arranged as

$$\mathbb{E}[R_i] = (1 - \beta)R_f + \beta(\mathbb{E}[R_m])$$

(replicating portfolio)

→ \$1 \$ -0.3 T-bill \$1.3 mk

Assume $\beta = 1.3$ for a stock. So, a \$1 invested in the asset will give the same average return if -\$0.30 is invested in the T-bills and \$1.30 invested in market (a levered position). CAPM's benchmark is a **replicating portfolio**.

- Factor benchmarks are combination of investible portfolios. Alpha is the excess return generated above the factor benchmark.

$$\mathbb{E}[R_i] = \alpha + (1 - \beta)R_f + \beta(\mathbb{E}[R_m])$$

Berkshire-Hathaway – Case Study



CAPM Summary

Alpha	0.465%
Beta	0.598
Adj R ²	0.23311

(Benchmark) { \$ 0.40 (T-Bill)
\$ 0.60 (Market)

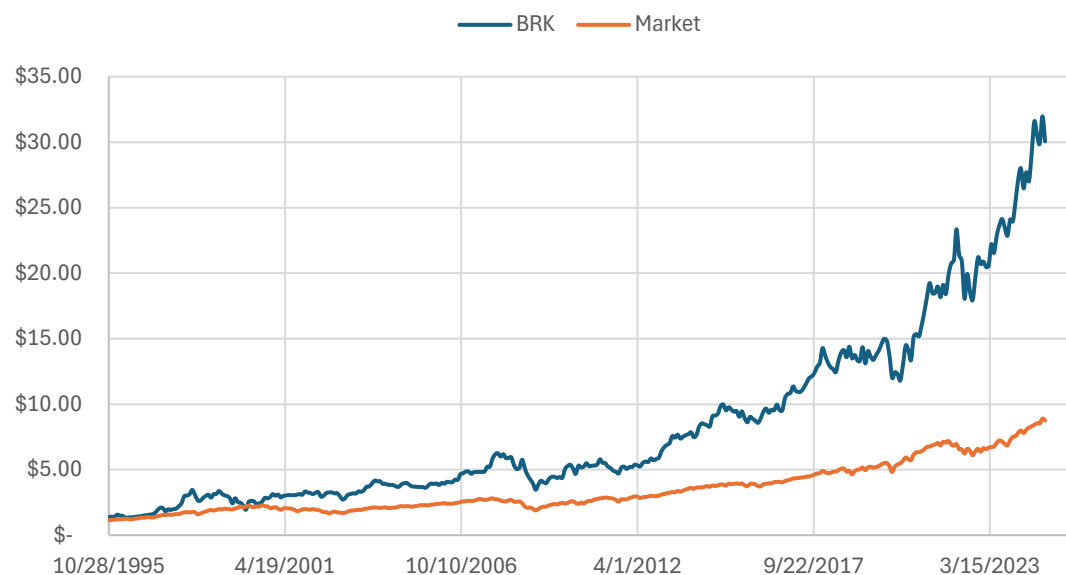
Buffet is adding **0.47%** extra return per month on top of the benchmark

(Fama-French Summary)

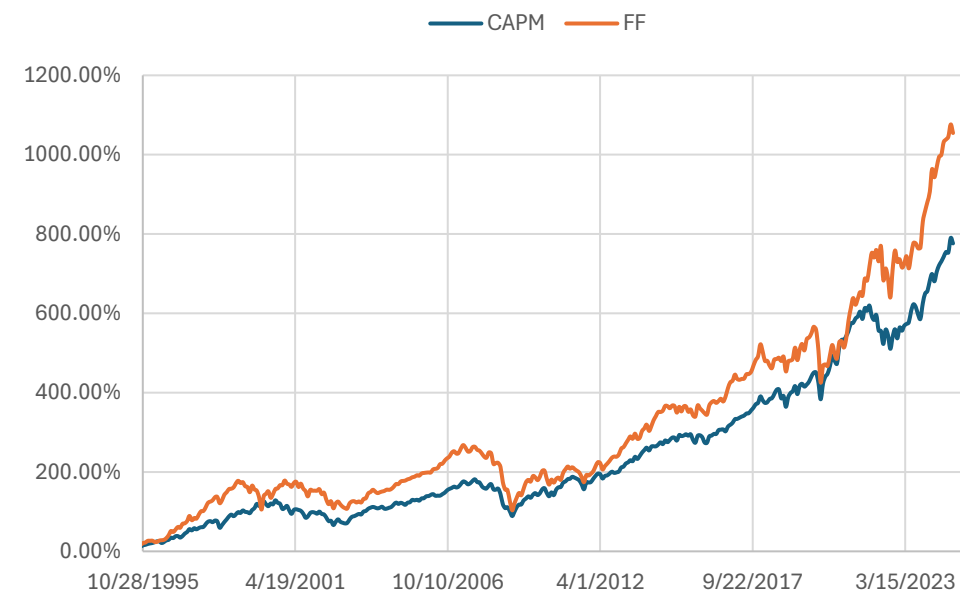
	Coefficients
Alpha	0.00358693
Market	0.71824063
SMB	-0.50701241
HML	0.44558448

(Benchmark) { 0.2817594 (T-Bill)
0.7182406 (Market)
-0.507012 (Small Cap)
0.5070124 (Large Cap)
0.4455845 (Value)
-0.445584 (Growth)

Cum gains between Berkshire Hathway vs Market



Factor Benchmark returns



Benchmark without risk free asset

- ▶ Benchmark need not have a risk-free asset. For example, CalPERS is one of the largest pension funds in US a benchmark can simply be a passive index of stocks and bonds.

$$R_{i,t} = \alpha + \beta_s R_{s,t} + \beta_b R_{b,t} + \epsilon_{i,t}$$

For a factor benchmark, we need $\alpha = 1$

$$\beta_s + \beta_b = 1$$

Comparison Case Study



► Below is the FF - regression coefficient estimates for 4 famous tickers under constant factor weights

②

	LSVEX	FMAGX	GSCGX	BRK
Alpha	0.00%	-0.27%	-0.14%	0.22%
t-stat	0.01	-2.23	-1.33	0.57
MKT Loading	0.94	1.12	1.04	0.36
t-stat	36.9	38.6	42.2	3.77
SMB Loading	0.01	-0.07	-0.12	-0.15
t-stat	0.21	-1.44	-3.05	-0.97
HML Loading	0.51	-0.05	-0.17	0.34
t-stat	14.6	-1.36	-4.95	2.57
UMD Loading	0.2	0.02	0.00	-0.06
t-stat	1.07	1.00	-0.17	-0.77

1. Which is the biggest value shop ? ✓
2. Which alpha is statistically significant ? ✓
3. Fidelity is a levered play in the market (T/F) ?
4. Which is a momentum player ?

$$(1 - \beta) \rightarrow \text{if } \beta \rightarrow \text{circled } \beta$$

Style Analysis – No shorting

- ▶ Sharpe introduced the idea of time varying benchmarks using a concept called as style analysis. Style analysis is a factor benchmark where the factor exposures evolve through time as opposed to having constant weights.
- ▶ Fama French factors are not directly tradeable. In style analysis, we can replicate a fund by using low-cost index ETFs (which are tradeable) and allow the regression coefficients to vary over time.
- ▶ For example, we can choose the ETFs below from the Spider family managed by State Street advisors.

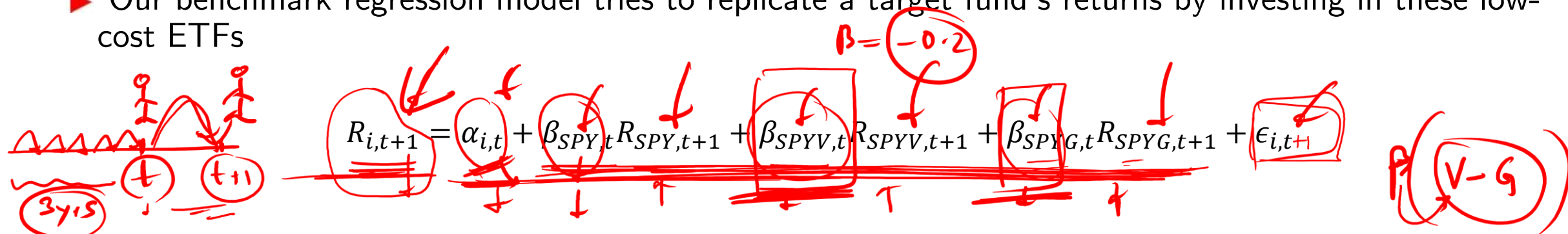
→ SPY: SPDR S&P 500 ETF, to mirror S&P 500; (Mkt)

→ SPYV: SPDR S&P 500 Value ETF, to mirror S&P 500 value index; (value)

→ SPYG: SPDR S&P 500 Growth ETF, to mirror S&P 500 growth index. (growth)

Style Analysis – No shorting

- Our benchmark regression model tries to replicate a target fund's returns by investing in these low-cost ETFs



$$R_{i,t+1} = \alpha_{i,t} + \beta_{SPY,t} R_{SPY,t+1} + \beta_{SPYV,t} R_{SPYV,t+1} + \beta_{SPYG,t} R_{SPYG,t+1} + \epsilon_{i,t+1}$$

With the constraint that $\beta_{SPY,t} + \beta_{SPYV,t} + \beta_{SPYG,t} = 1$ and $\beta_{SPY,t}, \beta_{SPYV,t}, \beta_{SPYG,t} \geq 0$

Note that the regression coefficients are estimated using the information up to time t . The return at time $t + 1$, is based on the replicating portfolio's return + fund's $\alpha_{i,t}$ and idiosyncratic component $\epsilon_{i,t}$

The excess return between period t to $t + 1$, is calculated as

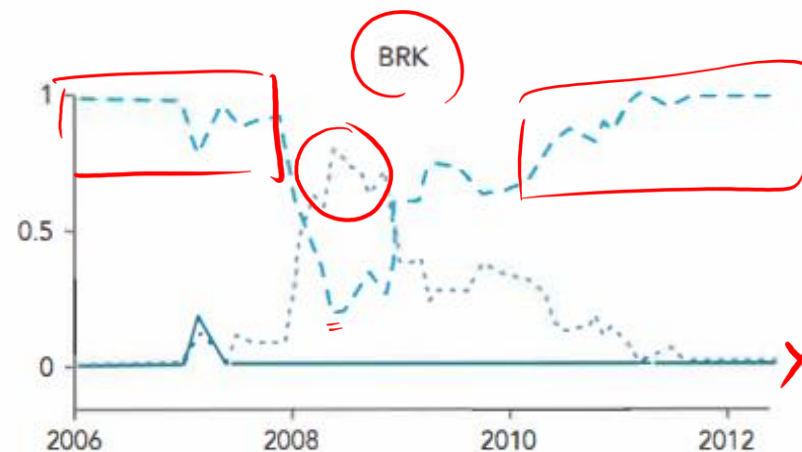
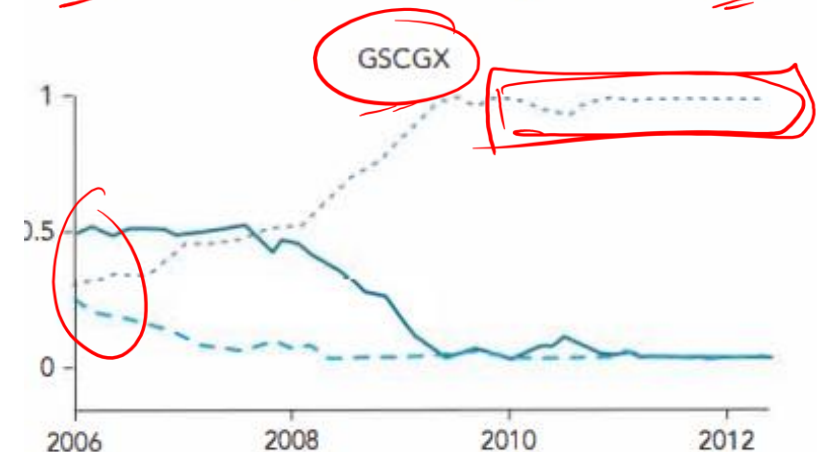
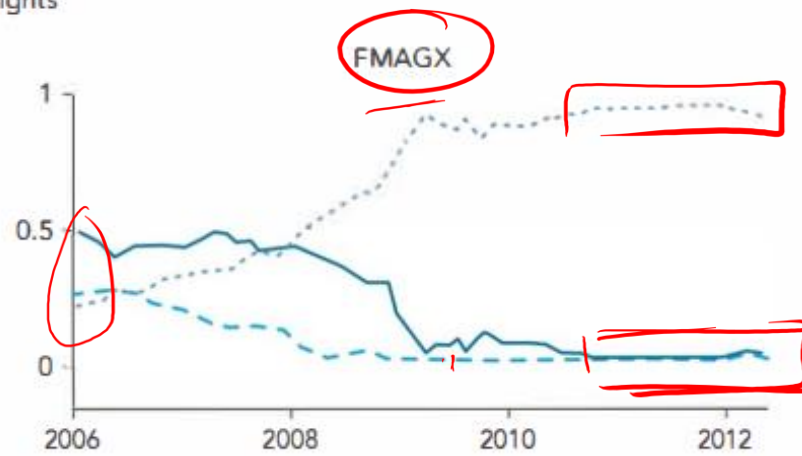


$$R_{i,t+1}^{ex} = R_{i,t+1} - \beta_{SPY,t} R_{SPY,t+1} + \beta_{SPYV,t} R_{SPYV,t+1} + \beta_{SPYG,t} R_{SPYG,t+1}$$

Comparison (factor-weights) – No Shorting

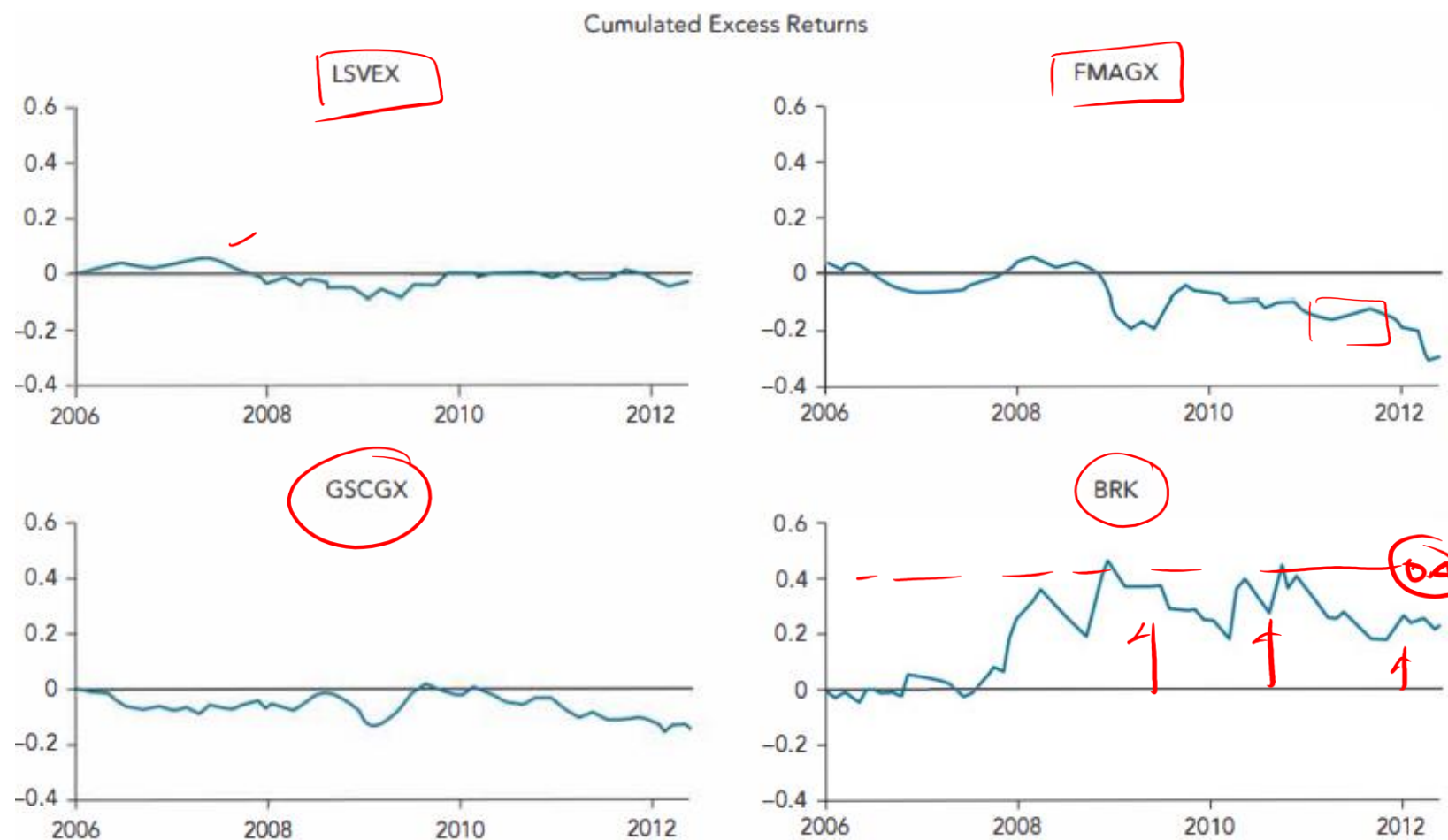
KALMAN

SPY SPYV SPYG
Factor Weights



1. LSVEX is only market and value tilt.
2. Fidelity and Goldman funds starts as a blend of all 3, but at the end it's only growth tilt.
3. Berkshire fund start with value to growth to value tilt.

Comparison (excess return) – No shorting



1. The only fund with excess avg return >0 is Berkshire fund

Style Analysis – with shorting

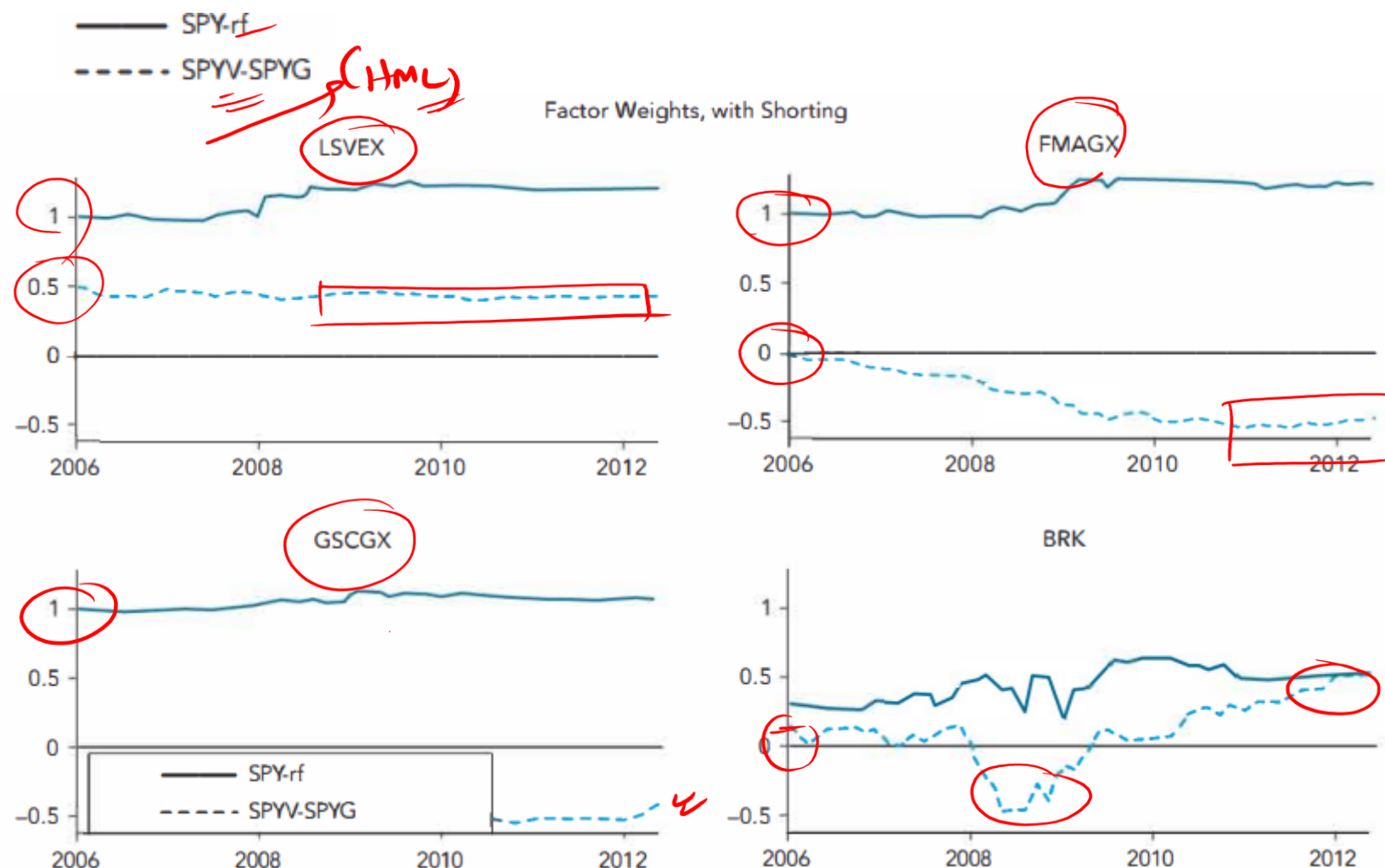
- If we allow investors to take short positions in ETFs, we can fit FF type model

$$R_{i,t+1} - R_{F,t+1} = \alpha_{i,t} + \beta_{SPY,t}(R_{SPY,t+1} - R_{F,t+1}) + h_t(R_{SPYV,t+1} - R_{SPYG,t+1}) + \epsilon_{i,t} \rightarrow$$

We do not need any constraints here as risk-free asset will absorb the residual weight to keep the total weight of benchmark = 1. Also, individual betas can be negative as shorting is allowed.

$$R_{i,t} = \alpha + R_f(1 - \beta_{SPY}) + \beta_{SPY} R_{SPY,t} + h_t R_{SPYV,t} - h_t R_{SPYG,t}$$

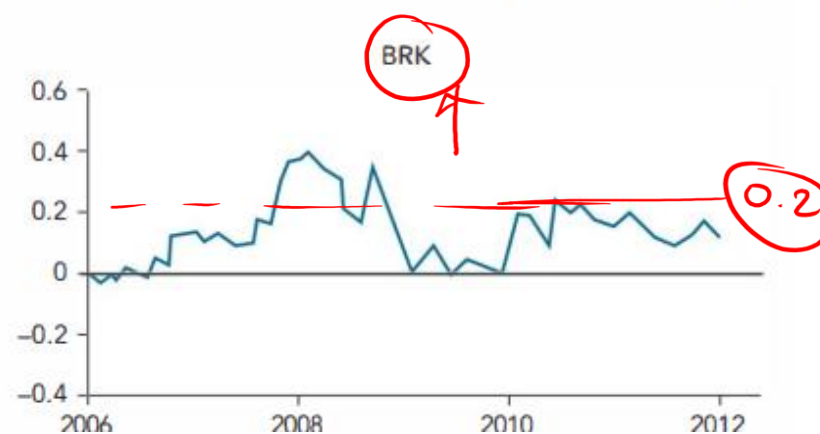
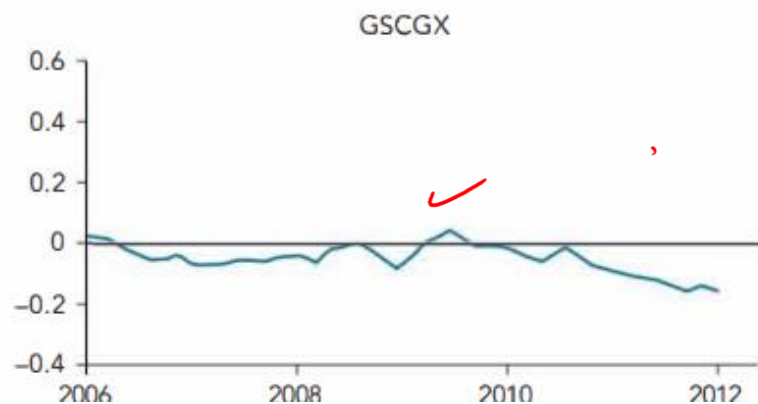
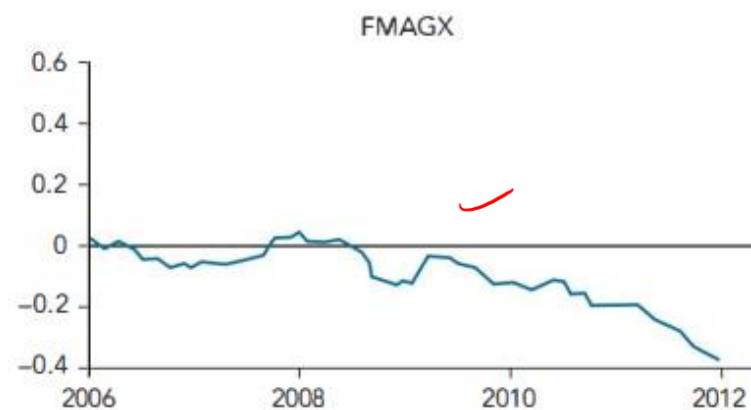
Comparison (factor-weights) – with shorting



1. LSVEX has a steady positive loading to VG factor.
2. Fidelity fund becomes more of a growth fund over time.
3. Berkshire fund start with value to growth to value tilt.

Comparison (excess return) – with shorting

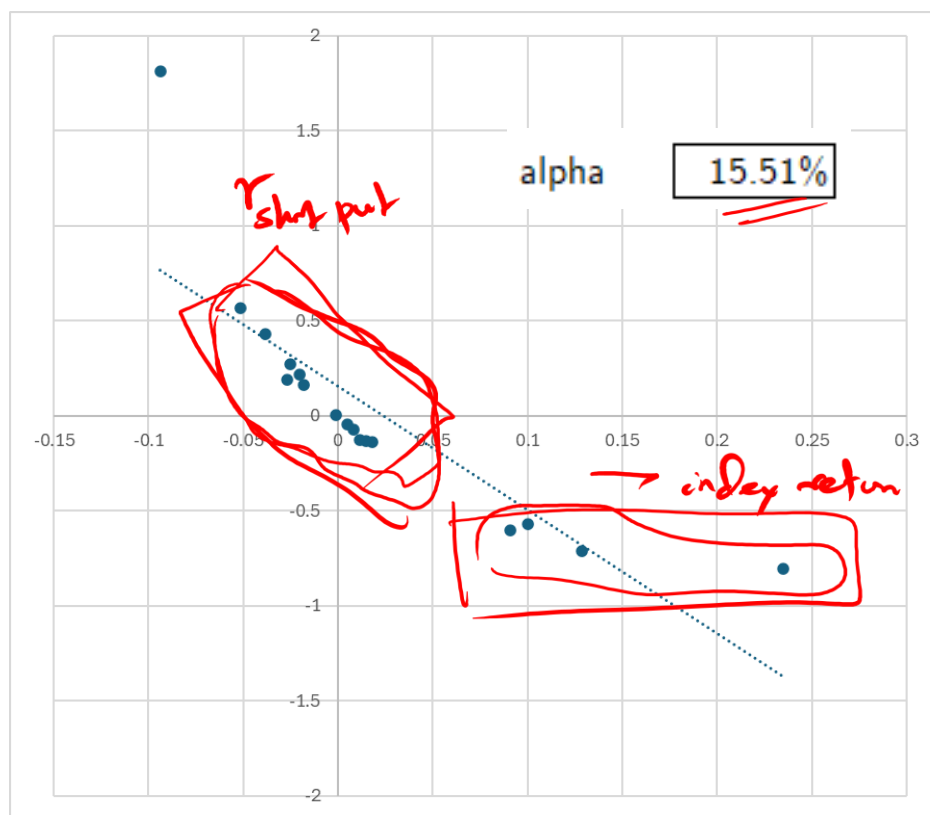
Cumulated Excess Returns, with Shorting



1. No major change to excess returns when we allow shorting in the replicating portfolio strategy
2. Alpha however reduces. (Why ?)

Non-linear payoffs

- Alpha, beta, Information ratio are all derived from a linear model. With non-linear strategies (such as options), alpha gets masked. Below example, shows that 'short put' position on market has an alpha of 15%. But we know that in BSM world, calls and puts do not create any additional value



- Short volatility strategies can inflate alpha. So, one remedy is to use either volatility factor or non-linear factors such as quadratic or max terms on the right-hand side

$$y = (\alpha) + \beta_1 x + \beta_2 x^2 + \beta_3 \max(x - k, 0) + \epsilon$$



Does alpha even exist ??

- ▶ Alpha is benchmark dependent.
- ▶ Literature suggests that it's a joint hypothesis problem where alpha and benchmark are simultaneously determined. It's always possible to find a benchmark ex-post that produces no alpha.